

Referring to right triangle BAC , we see that

$$\sin \phi = \frac{y}{10}$$

$$\text{or} \quad y = 10 \sin \phi = 10 \sin \left(\frac{1}{2}\pi - \theta\right).$$

Differentiating implicitly with respect to t and using the cofunction identity $\cos(\frac{1}{2}\pi - \theta) = \sin \theta$ yields

$$\frac{dy}{dt} = 10 \cos \left(\frac{1}{2}\pi - \theta\right) \left(0 - \frac{d\theta}{dt}\right) = -10 \sin \theta \frac{d\theta}{dt}.$$

(b) We must use radian measure for $d\theta/dt$. Since $15 = 15(\pi/180) = \pi/12$ radians, we substitute $d\theta/dt = \pi/12$ rad/hr and $\theta = 30^\circ = \pi/6$ in the formula for dy/dt , obtaining

$$\frac{dy}{dt} = -10 \left(\frac{1}{2}\right) \left(\frac{\pi}{12}\right) = -\frac{5\pi}{12} \approx -1.3 \text{ ft/hr.}$$

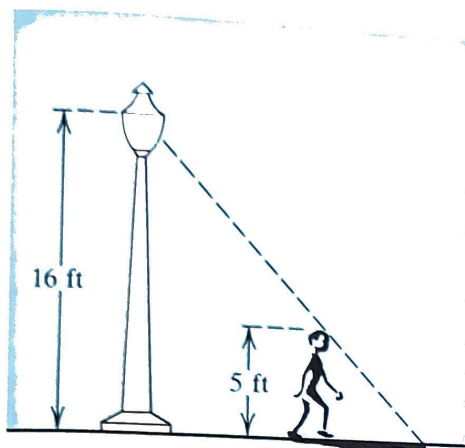
EXERCISES 3.8

Exer. 1–8: Assume that all variables are functions of t .

- 1 If $A = x^2$ and $\frac{dx}{dt} = 3$ when $x = 10$, find $\frac{dA}{dt}$.
- 2 If $S = z^3$ and $\frac{dz}{dt} = -2$ when $z = 3$, find $\frac{dS}{dt}$.
- 3 If $V = -5p^{3/2}$ and $\frac{dV}{dt} = -4$ when $V = -40$, find $\frac{dp}{dt}$.
- 4 If $P = 3/w$ and $\frac{dP}{dt} = 5$ when $P = 9$, find $\frac{dw}{dt}$.
- 5 If $x^2 + 3y^2 + 2y = 10$ and $\frac{dx}{dt} = 2$ when $x = 3$ and $y = -1$, find $\frac{dy}{dt}$.
- 6 If $2y^3 - x^2 + 4x = -10$ and $\frac{dy}{dt} = -3$ when $x = -2$ and $y = 1$, find $\frac{dx}{dt}$.
- 7 If $3x^2y + 2x = -32$ and $\frac{dy}{dt} = -4$ when $x = 2$ and $y = -3$, find $\frac{dx}{dt}$.
- 8 If $-x^2y^2 - 4y = -44$ and $\frac{dx}{dt} = 5$ when $x = -3$ and $y = 2$, find $\frac{dy}{dt}$.
- 9 As a circular metal griddle is being heated, its diameter changes at a rate of 0.01 cm/min. Find the rate at which the area of one side is changing when the diameter is 30 centimeters.
- 10 A fire has started in a dry, open field and spreads in the form of a circle. The radius of the circle increases at the rate of 6 ft/min. Find the rate at which the fire area is increasing when the radius is 150 feet.
- 11 Gas is being pumped into a spherical balloon at a rate of 5 ft³/min. Find the rate at which the radius is changing when the diameter is 18 inches.
- 12 Suppose a spherical snowball is melting and the radius is decreasing at a constant rate, changing from 12 inches to 8 inches in 45 minutes. How fast was the volume changing when the radius was 10 inches?
- 13 A ladder 20 feet long leans against a vertical building. If the bottom of the ladder slides away from the building horizontally at a rate of 3 ft/sec, how fast is the ladder sliding down the building when the top of the ladder is 8 feet from the ground?
- 14 A girl starts at a point A and runs east at a rate of 10 ft/sec. One minute later, another girl starts at A and runs north at a rate of 8 ft/sec. At what rate is the distance between them changing 1 minute after the second girl starts?
- 15 A light is at the top of a 16-foot pole. A boy 5 feet tall walks away from the pole at a rate of 4 ft/sec (see figure). At what rate is the tip of his shadow moving when he

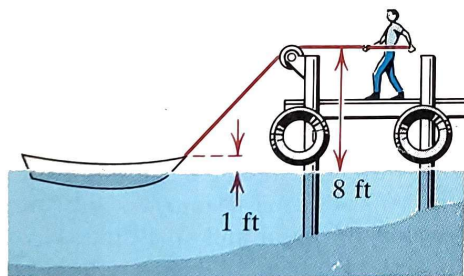
is 18 feet from the pole? At what rate is the length of his shadow increasing?

EXERCISE 15



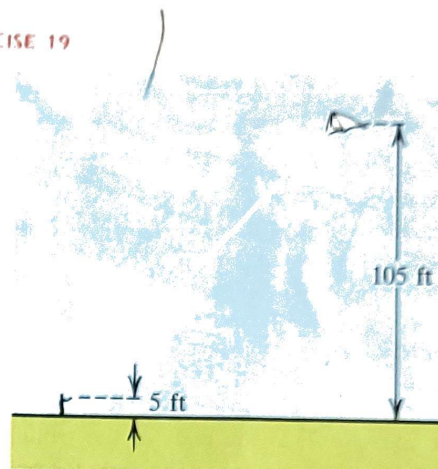
- 16 A man on a dock is pulling in a boat by means of a rope attached to the bow of the boat 1 foot above water level and passing through a simple pulley located on the dock 8 feet above water level (see figure). If he pulls in the rope at a rate of 2 ft/sec, how fast is the boat approaching the dock when the bow of the boat is 25 feet from a point that is directly below the pulley?

EXERCISE 16



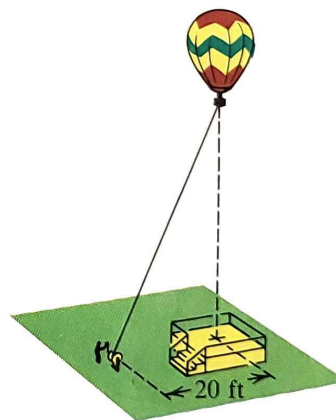
- 17 The top of a silo has the shape of a hemisphere of diameter 20 feet. If it is coated uniformly with a layer of ice and if the thickness is decreasing at a rate of $\frac{1}{4}$ in./hr, how fast is the volume of the ice changing when the ice is 2 inches thick?
- 18 As sand leaks out of a hole in a container, it forms a conical pile whose altitude is always the same as its radius. If the height of the pile is increasing at a rate of 6 in./min, find the rate at which the sand is leaking out when the altitude is 10 inches.
- 19 A person flying a kite holds the string 5 feet above ground level, and the string is payed out at a rate of 2 ft/sec as the kite moves horizontally at an altitude of 105 feet (see figure). Assuming there is no sag in the string, find the rate at which the kite is moving when 125 feet of string has been payed out.

EXERCISE 19



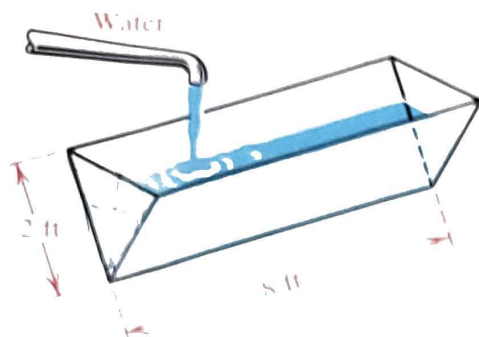
- 20 A hot-air balloon rises vertically as a rope attached to the base of the balloon is released at the rate of 5 ft/sec. The pulley that releases the rope is 20 feet from the platform where passengers board (see figure). At what rate is the balloon rising when 500 feet of rope has been payed out?

EXERCISE 20



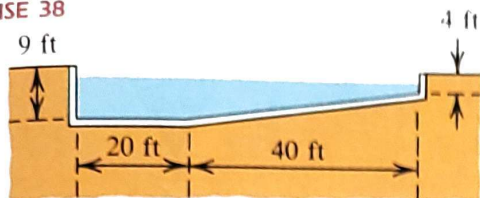
- 21 Boyle's law for confined gases states that if the temperature is constant, $p v = c$, where p is pressure, v is volume, and c is a constant. At a certain instant the volume is 75 in.³, the pressure is 30 lb/in.², and the pressure is decreasing at a rate of 2 lb/in.² every minute. At what rate is the volume changing at this instant?
- 22 A 100-foot-long cable of diameter 4 inches is submerged in seawater. Because of corrosion, the surface area of the cable decreases at a rate of 750 in.²/year. Ignoring the corrosion at the ends of the cable, find the rate at which the diameter is decreasing.
- 23 The ends of a water trough 8 feet long are equilateral triangles whose sides are 2 feet long (see figure on following page). If water is being pumped into the trough at a rate of 5 ft³/min, find the rate at which the water level is rising when the depth of the water is 8 inches.

EXERCISE 23



- 24 Work Exercise 23 if the ends of the trough have the shape of the graph of $y = 2|x|$ between the points $(-1, 2)$ and $(1, 2)$.
- 25 The area of an equilateral triangle is decreasing at a rate of $4 \text{ cm}^2/\text{min}$. Find the rate at which the length of a side is changing when the area of the triangle is 200 cm^2 .
- 26 Gas is escaping from a spherical balloon at a rate of $10 \text{ ft}^3/\text{hr}$. At what rate is the radius changing when the volume is 400 ft^3 ?
- 27 A stone is dropped into a lake, causing circular waves whose radii increase at a constant rate of 0.5 m/sec . At what rate is the circumference of a wave changing when its radius is 4 meters?
- 28 A softball diamond has the shape of a square with sides 60 feet long. If a player is running from second base to third at a speed of 24 ft/sec , at what rate is her distance from home plate changing when she is 20 feet from third?
- 29 When two resistors R_1 and R_2 are connected in parallel (see figure), the total resistance R is given by the equation $1/R = (1/R_1) + (1/R_2)$. If R_1 and R_2 are increasing at rates of 0.01 ohm/sec and 0.02 ohm/sec , respectively, at what rate is R changing at the instant that $R_1 = 30 \text{ ohms}$ and $R_2 = 90 \text{ ohms}$?
- EXERCISE 29**
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- 30 The formula for the adiabatic expansion of air is $pv^{1.4} = c$, where p is the pressure, v is the volume, and c is a constant. At a certain instant the pressure is 40 dyn/cm^2 and is increasing at a rate of 3 dyn/cm^2 per second. If, at that same instant, the volume is 60 cm^3 , find the rate at which the volume is changing.
- 31 If a spherical tank of radius a contains water that has a maximum depth h , then the volume V of water in the tank is given by $V = \frac{1}{3}\pi h^2(3a - h)$. Suppose a spherical tank of radius 16 feet is being filled at a rate of 100 gal/min . Approximate the rate at which the water level is rising when $h = 4$ feet ($1 \text{ gal} \approx 0.1337 \text{ ft}^3$).
- 32 A spherical water storage tank for a small community is coated uniformly with a 2-inch layer of ice. As the ice melts, the rate at which the volume of the ice decreases is directly proportional to the rate at which the surface area decreases. Show that the outside diameter is decreasing at a constant rate.
- 33 From the edge of a cliff that overlooks a lake 200 feet below, a boy drops a stone and then, two seconds later, drops another stone from exactly the same position. Discuss the rate at which the distance between the two stones is changing during the next second. (Assume that the distance an object falls in t seconds is $16t^2$ feet.)
- 34 A metal rod has the shape of a right circular cylinder. As it is being heated, its length is increasing at a rate of 0.005 cm/min and its diameter is increasing at 0.002 cm/min . At what rate is the volume changing when the rod has length 40 centimeters and diameter 3 centimeters?
- 35 An airplane is flying at a constant speed of 360 mi/hr and climbing at an angle of 45° . At the moment the plane's altitude is 10,560 feet, it passes directly over an air traffic control tower on the ground. Find the rate at which the airplane's distance from the tower is changing one minute later (neglect the height of the tower).
- 36 A North-South highway A and an East-West highway B intersect at a point P . At 10:00 A.M. an automobile crosses P traveling north on highway A at a speed of 50 mi/hr . At that same instant, an airplane flying east at a speed of 200 mi/hr and an altitude of 26,400 feet is directly above the point on highway B that is 100 miles west of P . If the airplane and the automobile maintain the same speed and direction, at what rate is the distance between them changing at 10:15 A.M.?
- 37 A paper cup containing water has the shape of a frustum of a right circular cone of altitude 6 inches and lower and upper base radii 1 inch and 2 inches, respectively. If water is leaking out of the cup at a rate of $3 \text{ in.}^3/\text{hr}$, at what rate is the water level decreasing when the depth of the water is 4 inches? (Note: The volume V of a frustum of a right circular cone of altitude h and base radii a and b is given by $V = \frac{1}{3}\pi h(a^2 + b^2 + ab)$.)
- 38 The top part of a swimming pool is a rectangle of length 60 feet and width 30 feet. The depth of the pool varies uniformly from 4 feet to 9 feet through a horizontal distance of 40 feet and then is level for the remaining 20 feet, as illustrated by the cross-sectional view in the figure. If the pool is being filled with water at a rate of 500 gal/min , approximate the rate at which the water

level is rising when the depth of the water at the deep end is 4 feet (1 gal ≈ 0.1337 ft³).

EXERCISE 38

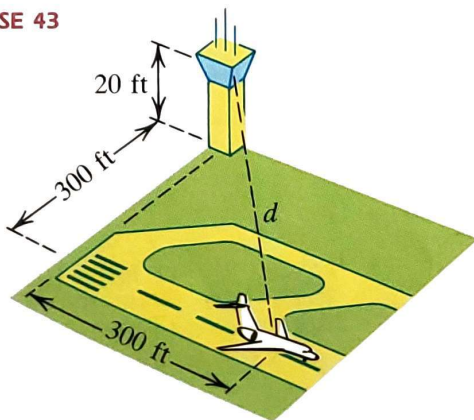
- 39 An airplane at an altitude of 10,000 feet is flying at a constant speed on a line that will take it directly over an observer on the ground. If, at a given instant, the observer notes that the angle of elevation of the airplane is 60° and is increasing at a rate of 1° per second, find the speed of the airplane.

- 40 In Exercise 16, let θ be the angle that the rope makes with the horizontal. Find the rate at which θ is changing at the instant that $\theta = 30^\circ$.

- 41 An isosceles triangle has equal sides 6 inches long. If the angle θ between the equal sides is changing at a rate of 2° per minute, how fast is the area of the triangle changing when $\theta = 30^\circ$?

- 42 A ladder 20 feet long leans against a vertical building. If the bottom of the ladder slides away from the building horizontally at a rate of 2 ft/sec, at what rate is the angle between the ladder and the ground changing when the top of the ladder is 12 feet above the ground?

- 43 The relative positions of an airport runway and a 20-foot-tall control tower are shown in the figure. The beginning of the runway is at a perpendicular distance of 300 feet from the base of the tower. If an airplane reaches a speed of 100 mi/hr after traveling 300 feet down the runway, at approximately what rate is the distance between the airplane and the control booth increasing at this time?

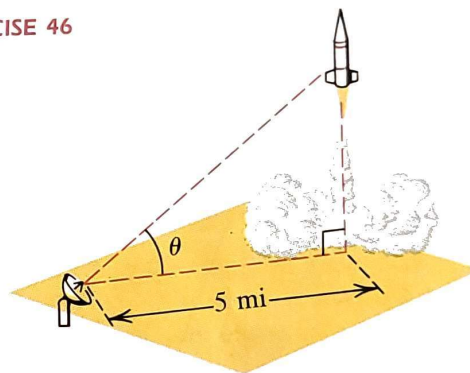
EXERCISE 43

- 44 The speed of sound in air at 0°C (or 273°K) is 1087 ft/sec, but this speed increases as the temperature rises. If T is temperature in $^\circ\text{K}$, the speed of sound v at this temper-

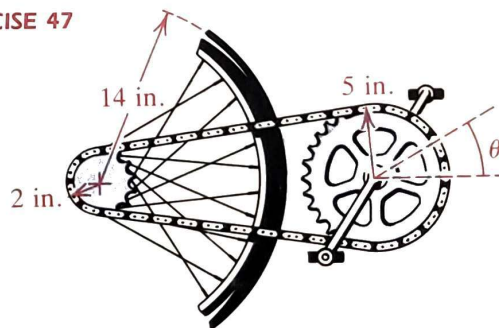
ature is given by $v = 1087\sqrt{T/273}$. If the temperature increases at the rate of 3°C per hour, approximate the rate at which the speed of sound is increasing when $T = 30^\circ\text{C}$ (or 303°K).

- 45 An airplane is flying at a constant speed and altitude, on a line that will take it directly over a radar station located on the ground. At the instant that the airplane is 60,000 feet from the station, an observer in the station notes that its angle of elevation is 30° and is increasing at a rate of 0.5° per second. Find the speed of the airplane.

- 46 A missile is fired vertically from a point that is 5 miles from a tracking station and at the same elevation (see figure). For the first 20 seconds of flight, its angle of elevation θ changes at a constant rate of 2° per second. Find the velocity of the missile when the angle of elevation is 30° .

EXERCISE 46

- 47 The sprocket assembly for a 28-inch bicycle is shown in the figure. Find the relationship between the angular velocity $d\theta/dt$ (in radians/sec) of the pedal assembly and the ground speed of the bicycle (in mi/hr).

EXERCISE 47

- 48 A 100-candlepower spotlight is located 20 feet above a stage (see figure on following page). The illuminance E (in footcandles) in the small lighted area of the stage is given by $E = (I \cos \theta)/s^2$, where I is the intensity of the light, s is the distance the light must travel, and θ is the indicated angle. As the spotlight is rotated through ϕ degrees, find the relationship between the rate of change in illumination dE/dt and the rate of rotation $d\phi/dt$.